ARCGIS INDOORS

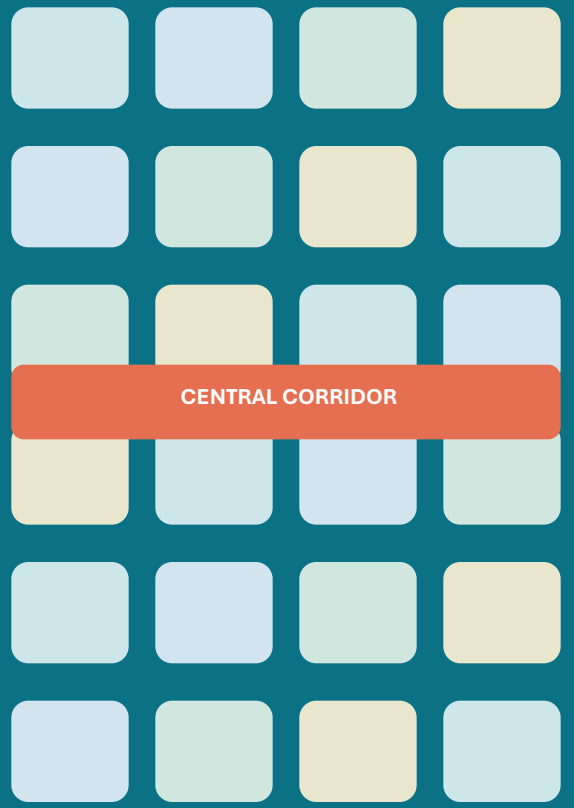
Asset enrichment and organizational catalog

A deterministic workflow for lifecycle history, procedural facilities, relationships, and optional local narratives.

EDITABLE POWERPOINT

VALIDATED CATALOG

Reporting date: October 1, 2026



Synthetic demonstration data

EXECUTIVE OUTCOME

A repeatable catalog, ready for demonstration

One reviewed facility is preserved; five facilities are generated without CAD or manual room inventories.

6
Sites and facilities

1,025
Units across 32 levels

4,852
Indoor lifecycle assets

0
Orphan references

What the workflow delivers

- Preserves Vertex Medical Center as a reviewed, enriched 1,003-asset slice.
- Creates four hospitals and Admin HQ with deterministic procedural floor plans.
- Builds validated lifecycle, inspection, repair, work-order, and dependency relationships.
- Uses local AI only for optional narrative text; Python remains authoritative.

DATA LINEAGE

One source, two controlled outputs

The original geodatabase is never updated in place.

01 READ-ONLY SOURCE

Vertex_Assets.gdb

Indoors hierarchy
1,003 indoor assets
History tables empty

02 REVIEWED VERTEX

Vertex_Assets_Enriched.gdb

Lifecycle planning
4,154 inspections
271 repairs | 501 work orders

03 SIX-FACILITY CATALOG

Organizational_Asset_Catalog.gdb

Vertex preserved
Five generated facilities
4,852 lifecycle assets

GlobalIDs are remapped inside each output; stable business identifiers preserve cross-build identity.

REVIEWED FACILITY

Vertex enrichment adds history, not source edits

Every planning value is synthetic, deterministic, and internally consistent.

1,003

enriched indoor assets

4,154	inspections
271	repairs
501	work orders

What is added

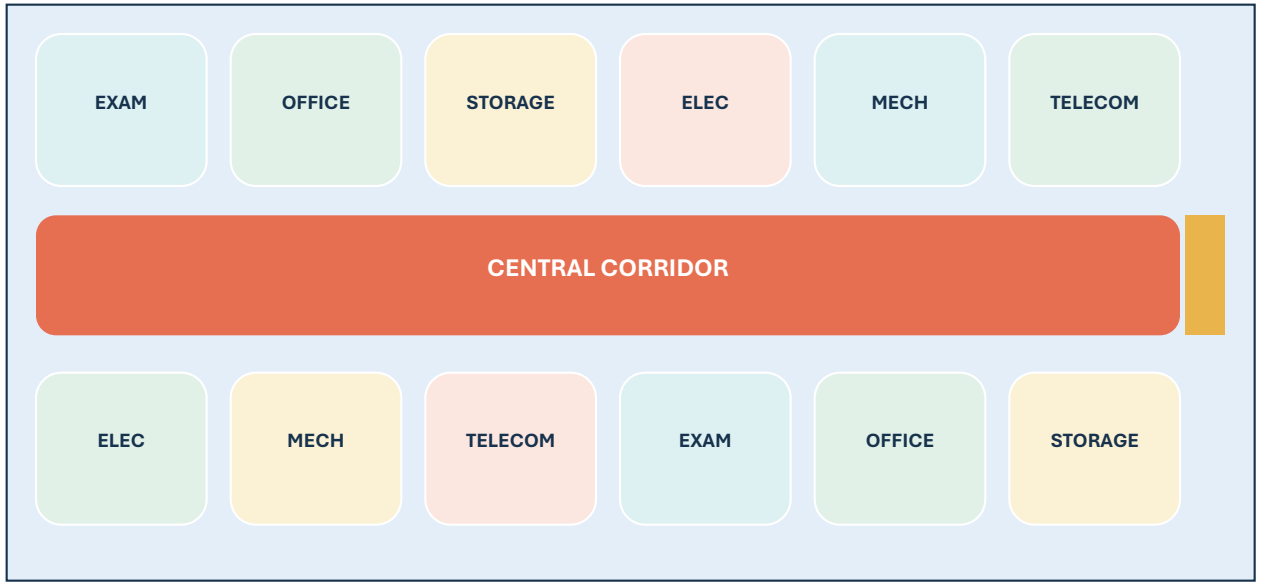
- Lifecycle stage, age, expected life, remaining life, and replacement year
- Inspection cadence, next maintenance, condition, criticality, and risk
- Purchase cost, replacement estimate, vendor, model, and warranty status
- Inspection, repair, and work-order histories with direct asset relationships

Source geometry and Indoors hierarchy remain unchanged.

Scale without CAD or manual room inventories

The floor-plan planner is deterministic, dependency-free, and profile-driven.

Representative generated level



Inputs

- Facility identity and seed
- Hospital or office profile
- Vertex spatial reference
- Synthetic geographic anchor

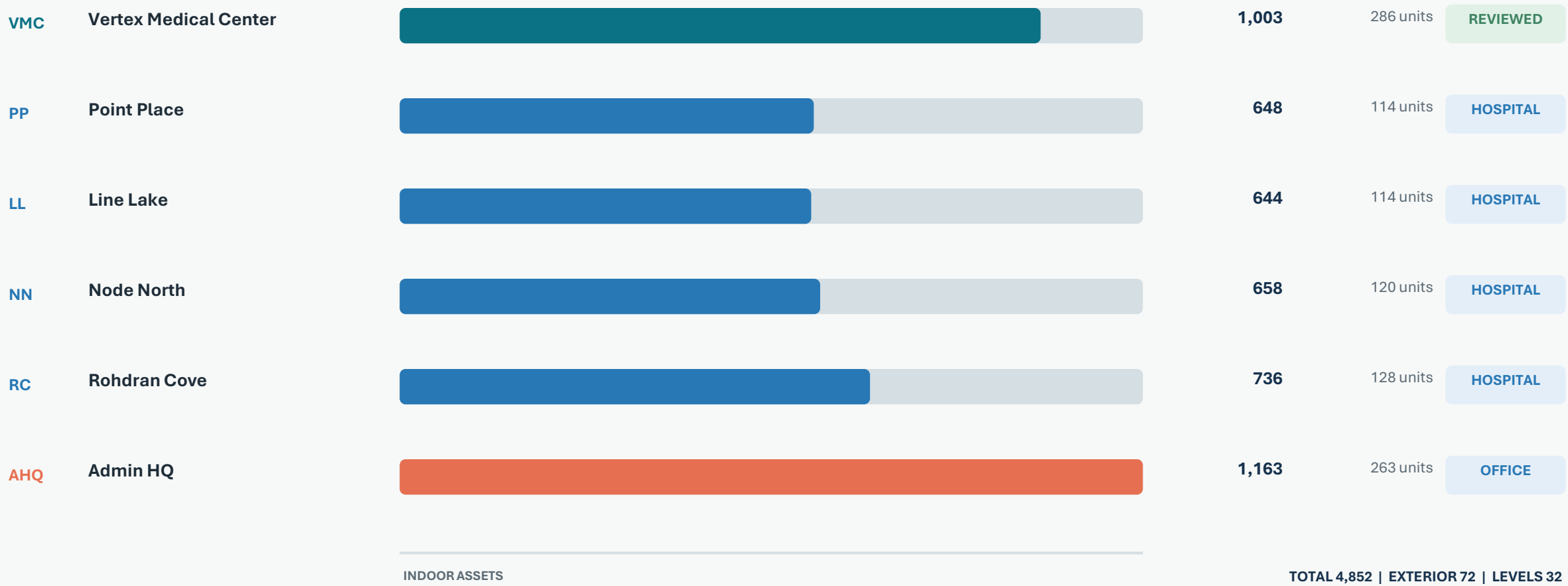
Outputs

Stable levels, rooms, names, use types, capacities, amenities, and asset allocation.

VALIDATED CATALOG

Six facilities form one comparable portfolio

Indoor asset counts vary with room programs rather than cloned floor plans.



TOTAL 4,852 | EXTERIOR 72 | LEVELS 32

PORTFOLIO PLANNING

Lifecycle distribution is exact by design

Facility mixes vary slightly while the organization-wide 20 / 50 / 30 contract remains exact.

4,852 lifecycle assets



970

Recently installed

At least 184 days; no more than 2 years or 25% of life

2,426

Within service life

Between 20% and 75% of expected service life

1,456

Retrofit / replacement

Between 80% and 105% of expected service life

DATA COVERAGE

The catalog supports planning and operations

Ten controlled asset classes span clinical, technology, safety, and building systems.

1. **Clinical + diagnostic**

   Patient-facing equipment, monitors, imaging, laboratory

2. **Building + electrical**

   HVAC, switchgear, plumbing, emergency power

3. **IT + communications**

   Workstations, network racks, conferencing, telecom

4. **Safety + security**

   Fire alarm, emergency lighting, cameras, access control

5. **Furniture**

   Clinical furniture, office furniture, storage

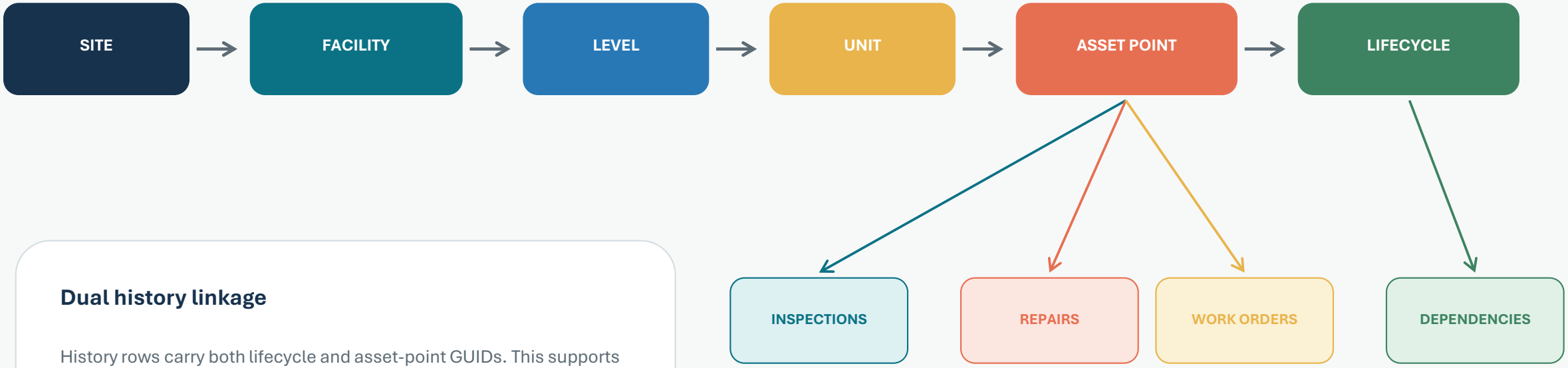
Planning fields

- Expected life: 4 to 30 years by class
- Replacement estimates: $250 to $350,000
- Condition: operational, maintenance due, offline repair
- Criticality: low through critical, rating 1 to 5
- Warranty, annual maintenance cost, overdue days, and risk
- Inspection cadence: typically 90 to 730 days

RELATIONSHIP ARCHITECTURE

Every asset remains connected to place and history

GlobalIDs and GUIDs are authoritative within each output; business IDs remain stable across rebuilds.



Dual history linkage

History rows carry both lifecycle and asset-point GUIDs. This supports lifecycle analysis and direct related records from floor-aware asset popups.

0 ORPHANS

3 DIRECT HISTORY RELATIONSHIPS

DETERMINISTIC + LLM BOUNDARY

Local AI adds wording, never authority

A strict localhost response contract and deterministic fallback keep every record complete.

DETERMINISTIC PYTHON

- IDs, hierarchy, and geometry
- Asset quantities and classes
- Dates, cohorts, conditions, and statuses
- Costs, warranty, maintenance, and risk
- Relationships and validation

OPTIONAL LOCAL NARRATOR

5,912
opportunities

500
requests sent

498
accepted

5,414
fallbacks

Contract localhost only | exactly one text field | 500 characters max

2 transport fallbacks | 5,412 budget fallbacks | 0 invalid responses

QUALITY CONTROLS

Validation makes the catalog demonstrably coherent

The manifest is the run audit record; tests protect deterministic behavior and promotion recovery.

0

orphan references

59 non-schema tests

11 notebook code cells compile

Validated contracts

- Exact Site, Facility, Level, Unit, and asset identities
- Complete parentage and relationship GUIDs
- Generated geometry containment and room-program continuity
- Inspection date bounds and latest-inspection consistency
- Exact lifecycle cohorts and imported Vertex preservation
- Lock-safe staged promotion with reusable validated recovery

RUN SEQUENCE

The operator path is controlled and reviewable

Production schema and catalog work runs in the ArcGIS Pro 3.7.2 notebook kernel.

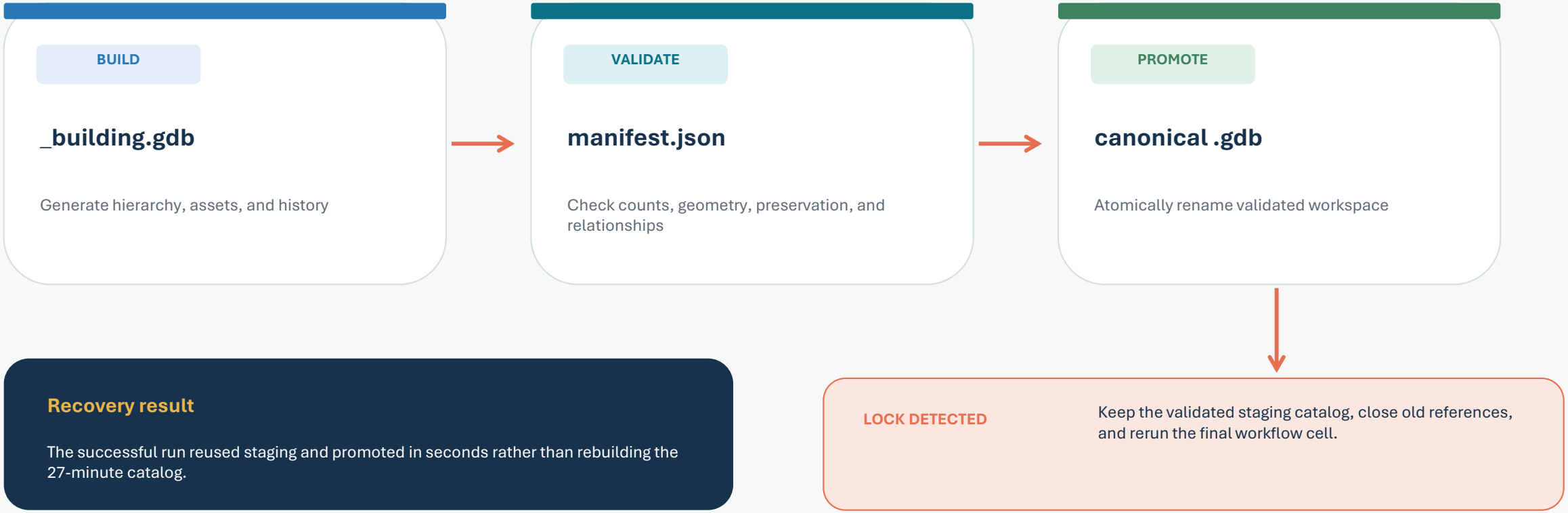
1. **Configure paths**
   Repository and source GDB
2. **Preflight**
   Environment, source, Ollama
3. **Enrich Vertex**
   Create separate output
4. **Review**
   Manifest and relationships
5. **Build catalog**
   Stage six facilities
6. **Validate**
   Counts, geometry, GUIDs
7. **Promote**
   Atomic rename to canonical

Close layers, standalone tables, attribute tables, and Catalog views that reference the old catalog before promotion.

PROMOTION AND RECOVERY

Staging protects the previous catalog

A validated build is reused after a lock failure; expensive generation does not repeat.



Recovery result

The successful run reused staging and promoted in seconds rather than rebuilding the 27-minute catalog.

APPLICATION PATTERNS

One catalog supports several demonstration stories

Publish selected layers and relationship classes together for the richest experience.

PORTFOLIO

Dashboards

Compare facilities, lifecycle stages, condition, cost, and risk.

FLOOR-AWARE

Related records

Open inspections, repairs, and work orders from an indoor asset popup.

CONNECTED

Knowledge graph

Explore assets, locations, lifecycle history, and supporting systems.

REPEATABLE

Training + testing

Use stable synthetic data without production credentials or enterprise IDs.

SCOPE

Clear boundaries keep the demonstration honest

The catalog is designed for evaluation and storytelling, not as an asserted facility inventory.

IN SCOPE

- Synthetic lifecycle and history records
- Procedural hospital and office layouts
- Dashboard and related-record workflows
- Replacement, warranty, risk, and maintenance planning
- Graph-ready system dependencies

OUT OF SCOPE

- Claims about real equipment inventory
- Reconstruction of actual building plans
- Utility-network asset modeling
- Live enterprise records or credentials
- Hosted forms, portal IDs, or image binaries

READY FOR DEMONSTRATION

A coherent catalog with room to become a product

Next actions

- Publish asset and history layers with relationship classes
- Configure floor awareness and related-record popups
- Build portfolio and replacement-planning dashboards
- Model system dependencies as graph relationships
- Decide whether enrichment becomes a toolbox tool

Validated result

6	facilities
1,025	units
4,852	assets
0	orphans

Documentation and source

github.com/JonGabHall/demo-data-generator